

Revised Load Interconnection System Impact Study Report

For Queue Project Q15-0528

Hebron Rosemeade 100 kV
November 26, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0528. The Interconnection Customer proposes a 164.1 MW large load (data center) facility interconnecting at the Hebron Rosemeade 100 kV substation in the Indiana area. The request was received on October 22, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 184 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,497,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Hart / Portage Uppco 100 kV
Contingency: Waynedale / Monfort 230 kV
Loading: 121.9% (Rating: 126 MVA)
DFAX: 14.4%
Pre-project loading: 103.2%
2. Facility: Southport / Hart 100 kV
Contingency: Monfort / Waynedale 230 kV
Loading: 113.4% (Rating: 117 MVA)
DFAX: 14.4%
Pre-project loading: 93.2%
3. Facility: Illinois Road / Southport Substation 100 kV
Contingency: Waynedale / Monfort 230 kV
Loading: 108.9% (Rating: 102 MVA)
DFAX: 14.4%
Pre-project loading: 85.7%
4. Facility: Hillcrest / Wayne Trace 100 kV
Contingency: Allen / Aep Pettit Avenue 100 kV
Loading: 102.6% (Rating: 153 MVA)
DFAX: 5.3%
Pre-project loading: 96.9%
5. Facility: Portage Uppco / Columbia Hills 100 kV Line
Contingency: Hart / Portage Uppco 100 kV line
Loading: 100.6% (Rating: 123 MVA)
DFAX: 7.9%
Pre-project loading: 90.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hart / Portage Uppco 100 kV	Rebuild line with higher-capacity conductor	\$4,331,000	26% (\$1,115,000)
Southport / Hart 100 kV	Rebuild line with higher-capacity conductor	\$2,986,000	26% (\$769,000)
Illinois Road / Southport 100 kV	Reconductor line	\$1,592,000	26% (\$410,000)
Hillcrest / Wayne Trace Substation 100 kV L	Reconductor line	\$573,000	3% (\$20,000)
Columbia Hills / Portage Uppco 100 kV	Reconductor line	\$1,419,000	13% (\$183,000)
	Total allocated to project		\$2,497,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.